

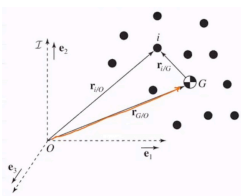
position of particle $j$ in inertial frame ${}^{I}\vec{r}_{j/o}$

total force on particle $j$: $\vec{F}_j = \vec{F}_{j,exe} + \sum_{i \neq j} \vec{F}_{j,i}$

external force on $j$ — force on $j$ due to $i$

$m_j \vec{a}_{j/o} = \vec{F}_j = \vec{F}_{j,exe} + \sum_{i \neq j} \vec{F}_{j,i}$

## · Center of Mass



Center of mass: ${}^{I}\vec{r}_{G/o} = \frac{1}{m_a} \sum_i m_i {}^{I}\vec{r}_{i/a}$ $\quad (m_a = \sum_i m_i)$

Center of mass corollary: $\sum_i m_i {}^{I}\vec{r}_{i/a} = 0$

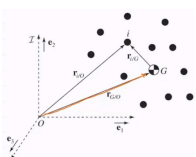
$${}^{I}\vec{r}_{a/o} = \frac{1}{m_a} \sum_i m_i {}^{I}\vec{r}_{i/o}$$

$$m_a {}^{I}\vec{r}_{a/o} = \sum_i m_i ({}^{I}\vec{r}_{i/a} + {}^{I}\vec{r}_{a/o}) = m_a {}^{I}\vec{r}_{a/o} + \sum_i m_i {}^{I}\vec{r}_{i/a}$$

$$\vec{F}_{exe} = m_a {}^{I}\vec{a}_{a/o}$$

$${}^{I}\vec{a}_{a/o} = \frac{1}{m_a} \sum_i m_i {}^{I}\vec{a}_{i/o} = \frac{1}{m_a} \sum_i \left[ \vec{F}_{i,exe} + \sum_{j \neq i} \vec{F}_{j,i} \right] = \frac{1}{m_a} \sum_i \vec{F}_{i,exe} = \frac{1}{m_a} \vec{F}_{exe}$$

## · Total Linear Momentum



linear momentum of $i$ wrt $o$

${}^{I}\vec{P}_{i/o} = m_i {}^{I}\vec{V}_{i/o}$

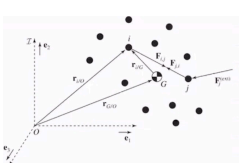
total linear momentum

${}^{I}\vec{P}_{s/o} = \sum_i m_i {}^{I}\vec{V}_{i/o} = m_a \frac{1}{m_a} \sum_i m_i {}^{I}\vec{V}_{i/o} = m_a {}^{I}\vec{V}_{a/o} = {}^{I}\vec{P}_{a/o}$

$s$ stands for "whole system"

$\frac{d}{dt} {}^{I}\vec{P}_{s/o} = \frac{d}{dt} {}^{I}\vec{P}_{a/o} = m_a {}^{I}\vec{a}_{a/o} = \vec{F}_{exe}$ $\qquad$ If $\vec{F}_{exe} = 0$, total linear momentum is conserved

## · Total Angular Momentum



Angular momentum of $i$ wrt $o$

${}^{I}\vec{h}_{i/o} = m_i {}^{I}\vec{r}_{i/o} \times {}^{I}\vec{V}_{i/o}$

$$\frac{d}{dt} {}^{I}\vec{h}_{i/o} = m_i \left[ {}^{I}\vec{V}_{i/o} \times {}^{I}\vec{V}_{i/o} + {}^{I}\vec{r}_{i/o} \times {}^{I}\vec{a}_{i/o} \right]$$

$$= {}^{I}\vec{r}_{i/o} \times \vec{F}_i$$

$$= \underbrace{{}^{I}\vec{r}_{i/o} \times \vec{F}_{i,exe}}_{\text{external moment}} + \underbrace{{}^{I}\vec{r}_{i/o} \times \sum_j \vec{F}_{i,j}}_{\text{internal moment}}$$

## Total Angular Momentum

$$\underbrace{{}^I\vec{h}_{S/O}}_{} = \sum_i {}^I\vec{h}_{i/O} = \sum_i m_i {}^I\vec{r}_{i/O} \times {}^I\vec{v}_{i/O}$$

S stands for "whole system"

$$\frac{d}{dt}{}^I\vec{h}_{S/O} = \sum_i \left[ {}^I\vec{r}_{i/O} \times \vec{F}_{i,ext} + {}^I\vec{r}_{i/O} \times \sum_j \vec{F}_{ij} \right]$$

$$= \sum_i {}^I\vec{r}_{i/O} \times \vec{F}_{i,ext}$$

o under internal moment assumption: $\vec{F}_{ij}$ parallel with $\vec{r}_{ij}$

If no external torque, total angular momentum is conserved

$$\sum_i {}^I\vec{r}_{i/O} \times \sum_j \vec{F}_{ij} = \sum_i \sum_j {}^I\vec{r}_{i/O} \times \vec{F}_{ij}$$

$$= \frac{1}{2}\left[ \sum_i \sum_j {}^I\vec{r}_{i/O} \times \vec{F}_{ij} + \sum_j \sum_i {}^I\vec{r}_{j/O} \times \vec{F}_{ji} \right]$$

$$= \frac{1}{2} \sum_i \sum_j \left( {}^I\vec{r}_{i/O} - {}^I\vec{r}_{j/O} \right) \times \vec{F}_{ij}$$

$$= 0$$

## Angular momentum Separation / Decomposition

$${}^I\vec{h}_{S/O} = {}^I\vec{h}_{A/O} + {}^I\vec{h}_{S/A}$$

$${}^I\vec{h}_{S/O} = \sum_i m_i {}^I\vec{r}_{i/O} \times {}^I\vec{v}_{i/O}$$

$$= \sum_i m_i \left( {}^I\vec{r}_{i/A} + {}^I\vec{r}_{A/O} \right) \times \left( {}^I\vec{v}_{i/A} \times {}^I\vec{v}_{A/O} \right)$$

$$= \sum_i m_i {}^I\vec{r}_{i/A} \times {}^I\vec{v}_{i/A} + \underbrace{\sum_i m_i {}^I\vec{r}_{i/A}}_{0} \times {}^I\vec{v}_{A/O} + \underbrace{\sum_i m_i {}^I\vec{r}_{A/O} \times {}^I\vec{v}_{i/A}}_{0} + \sum_i m_i {}^I\vec{r}_{A/O} \times {}^I\vec{v}_{A/O}$$

$$= {}^I\vec{h}_{S/A} + {}^I\vec{h}_{A/O}$$

## Total kinetic Energy

Individual kinetic energy: $T_{i/O} = \frac{1}{2} m_i \| {}^I\vec{v}_{i/O} \|^2$

total kinetic energy: $\underbrace{T_{S/O}}_{} = \sum_i \frac{1}{2} m_i \| {}^I\vec{v}_{i/O} \|^2$

S stands for "whole system"

## Kinetic enagy separation / decomposition

$$T_{S/O} = T_{A/O} + T_{S/A}$$

$$T_{S/O} = \sum_i \frac{1}{2} m_i \| {}^I\vec{v}_{i/A} + {}^I\vec{v}_{A/O} \|^2$$

$$= \sum_i \frac{1}{2} m_i \| {}^I\vec{v}_{i/A} \|^2 + \sum_i \frac{1}{2} m_i \| {}^I\vec{v}_{A/O} \|^2 + \sum_i m_i \langle {}^I\vec{v}_{i/A}, {}^I\vec{v}_{A/O} \rangle$$

$$= \sum_i \frac{1}{2} m_i \| {}^I\vec{v}_{i/A} \|^2 + \frac{1}{2} m_A \| {}^I\vec{v}_{A/O} \|^2$$

## Moment of Inertia

$${}^I\vec{h}_{S/A} = \sum_i m_i {}^I\vec{r}_{i/A} \times {}^I\vec{v}_{i/A}$$

$$= \sum_i m_i {}^I\vec{r}_{i/A} \times \left[ \left( {}^B\vec{v}_{i/A} \right)_{fr} + \omega_{B/F} \times {}^I\vec{r}_{i/A} \right]$$

$$= \sum_i m_i {}^I\vec{r}_{i/A} \times \omega_{B/F} \times {}^I\vec{r}_{i/A}$$

$$\left( {}^I\vec{h}_{S/A} \right)_B = \sum_i m_i {}^B\vec{r}_{i/A} \times \omega_{B/F} \times {}^B\vec{r}_{i/A}$$

$$= -\sum_i m_i \left[ {}^B r_{i/A} \right] \left[ {}^B r_{i/A} \right] \omega_{B/F}$$

$$= \underbrace{\sum_i m_i \left[ {}^B r_{i/A} \right]^T \left[ {}^B r_{i/A} \right]}_{} \omega_{B/F}$$

${}^B I_A$: Moment of Inertia

$$\begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix}$$

$$[r][r] = \begin{bmatrix} 0 & -r_3 & r_2 \\ r_3 & 0 & -r_1 \\ -r_2 & r_1 & 0 \end{bmatrix} \begin{bmatrix} -r_2^2 - r_3^2 & r_1 r_2 & r_1 r_3 \\ r_2 r_1 & -r_3^2 - r_1^2 & r_2 r_3 \\ r_3 r_1 & r_3 r_2 & -r_1^2 - r_2^2 \end{bmatrix}$$

$$= -\|r\|_2^2 I + r r^T$$

$${}^B I_A = \sum_i m_i \left[ {}^B r_{i/A} \right]^T \left[ {}^B r_{i/A} \right]$$

$$= \sum_i m_i \begin{bmatrix} r_2^2 + r_3^2 & -r_1 r_2 & -r_1 r_3 \\ -r_2 r_1 & r_1^2 + r_3^2 & -r_2 r_3 \\ -r_3 r_1 & -r_3 r_2 & r_1^2 + r_2^2 \end{bmatrix} \Bigg|_{r = {}^B r_{i/A}}$$

$({}^{I}h_{s/a})_{\mathcal{B}} = {}^{\mathcal{B}}I_a \, {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}}$

Total kinetic Energy

$$\begin{aligned} T_{s/o} &= \sum \tfrac{1}{2} m_i \| {}^{I}\vec{V}_{i/o} \|^2 \\ &= \sum \tfrac{1}{2} m_i \| {}^{I}\vec{V}_{i/a} \|^2 + \sum \tfrac{1}{2} m_i \| \vec{V}_{a/o} \|^2 \\ &= \sum \tfrac{1}{2} m_i \| ({}^{\mathcal{B}}\vec{V}_{i/a})_{\mathcal{I}} + {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{I}} \times {}^{I}\vec{r}_{i/a} \|^2 + T_{a/o} \\ &= \sum \tfrac{1}{2} m_i \| {}^{\mathcal{B}}\vec{V}_{i/a} \times {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}} \|^2 + T_{a/o} \\ &= \sum \tfrac{1}{2} m_i \, {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}}^{T} [{}^{\mathcal{B}}\vec{r}_{i/a}]^{T} [{}^{\mathcal{B}}\vec{r}_{i/a}] \, {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}} + T_{a/o} \end{aligned}$$

$$T_{s/o} = T_{s/a} + T_{a/o} = \tfrac{1}{2} \, {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}}^{T} \, {}^{\mathcal{B}}I_a \, {}^{\mathcal{B}}\omega_{\mathcal{B}/\mathcal{B}} + T_{a/o}$$

Parallel axis theorem

$${}^{\mathcal{B}}I_o = {}^{\mathcal{B}}I_a + m_a \left[ \| {}^{\mathcal{B}}r_{a/o} \|^2 I - {}^{\mathcal{B}}r_{a/o} \, {}^{\mathcal{B}}r_{a/o}^{T} \right]$$

$$\begin{aligned} {}^{\mathcal{B}}I_o &= -\sum m_i [{}^{\mathcal{B}}r_{i/o}][{}^{\mathcal{B}}r_{i/o}] \\ &= -\sum m_i [{}^{\mathcal{B}}r_{i/a} + {}^{\mathcal{B}}r_{a/o}][{}^{\mathcal{B}}r_{i/a} + {}^{\mathcal{B}}r_{a/o}] \\ &= -\sum m_i [{}^{\mathcal{B}}r_{i/a}][{}^{\mathcal{B}}r_{i/a}] - \sum m_i [{}^{\mathcal{B}}r_{a/o}][{}^{\mathcal{B}}r_{a/o}] \\ &= {}^{\mathcal{B}}I_a + m_a \left[ \| {}^{\mathcal{B}}r_{a/o} \|^2 I - {}^{\mathcal{B}}r_{a/o} \, {}^{\mathcal{B}}r_{a/o}^{T} \right] \end{aligned}$$

- Continuous Case

$${}^{\mathcal{B}}I_a = \iiint \rho(x,y,z) \begin{bmatrix} y^2+z^2 & -xy & -xz \\ -yz & x^2+z^2 & -yz \\ -zx & -zy & x^2+y^2 \end{bmatrix} dx\,dy\,dz$$

- Properties

① ${}^{\mathcal{B}}I_a$ is strictly positive-definite (for non-string object)

$${}^{\mathcal{B}}I_a = -\sum m_i [{}^{\mathcal{B}}r_{i/a}][{}^{\mathcal{B}}r_{i/a}] = \sum m_i \left( \| {}^{\mathcal{B}}r_{i/a} \|^2 I - {}^{\mathcal{B}}r_{i/a} \, {}^{\mathcal{B}}r_{i/a}^{T} \right)$$

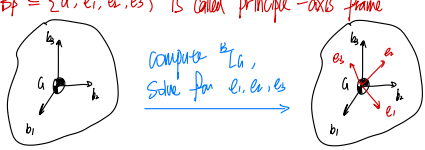
$\| r \|^2 I - r r^T \geq 0$, the only nullspace is along $r$ direction

then the only case ${}^{\mathcal{B}}I_a$ is not PD is that the object is a string

then all $\| r \|^2 I - r r^T$ have nullspace in direction of the string

② Let the eigenvalue-eigenvector pair of ${}^{\mathcal{B}}I_a$ be $(\lambda_1, e_1)$ $(\lambda_2, e_2)$ $(\lambda_3, e_3)$

$\mathcal{B}_p = \{a, e_1, e_2, e_3\}$ is called principle-axis frame




$\mathcal{B}_p$ might be easier to use, but need to reorder $(e_1, e_2, e_3)$ to follow right-hand rule

Can decompose $\omega_{\mathcal{B}/\mathcal{B}}$ in $\{e_1, e_2, e_3\}$

$$\left(\vec{\tau}_{S/A}\right)_{/B} = {}^{B}I_A \, W_{B/B}$$

$$= \begin{bmatrix} | & | & | \\ e_1 & e_2 & e_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} - & e_1^T & - \\ - & e_2^T & - \\ - & e_3^T & - \end{bmatrix} (W_{B/B})$$

$$= \begin{bmatrix} | & | & | \\ e_1 & e_2 & e_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} \begin{bmatrix} - & e_1^T & - \\ - & e_2^T & - \\ - & e_3^T & - \end{bmatrix} \begin{bmatrix} | & | & | \\ e_1 & e_2 & e_3 \\ | & | & | \end{bmatrix} (W_{B/B})$$

$$= \begin{bmatrix} | & | & | \\ e_1 & e_2 & e_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} (W_{B/B})$$

$$\left(\vec{\tau}_{S/A}\right)_{/ep} = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix} (W_{B/B})$$

- Moments

0-th moment: how much $\quad m_A = \iiint \rho \, dv \quad \in \mathbb{R}$

1st moment: where $\quad r_{A/O} = \iiint \rho \cdot r \, dv \quad \in \mathbb{R}^3$

2nd moment: how spreaded $\quad {}^{B}I_A = \iiint \rho \, [r]^T [r] \, dv \quad \in \mathbb{R}^9$